What Journal Statistical Editors are Looking For

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- National Institutes of Health Funding
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- American Headache Society
  - Statistical Editor*, Headache*

Statistical Peer Review

"…the majority of statistical analyses are performed by people with an inadequate understanding of statistical methods. They are then peer reviewed by people who are generally no more knowledgeable"

-Douglas Altman

Journal Statistical Editors

- Provide ‘specialty’ peer review
  - Research methods
  - Measurement theory
  - Statistical analysis

- In conjunction with handling editors and editor in chief:
  - Evaluate suitability of research design
  - Evaluate statistical analysis plan
  - Evaluate completeness of reporting
  - Assess probability of replication

What Journal Editors Are Trying To ***Avoid***:

Piller (2022).
Science, 377,
Issue 6604

nature Vol 440|16 March 2006|**doi:10.1038/nature04533**

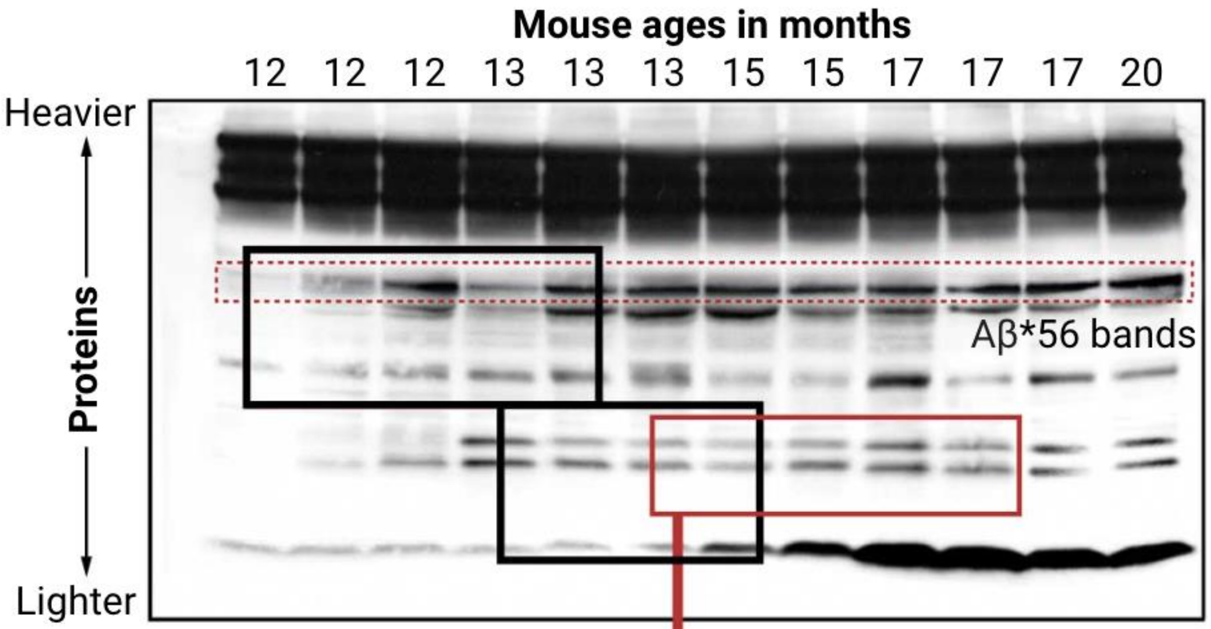
LETTERS

A specific amyloid-β protein assembly in the brain impairs memory

Sylvain Lesné[1], Ming Teng Koh[4], Linda Kotilinek[1], Rakez Kayed[6], Charles G. Glabe[6], Austin Yang[7], Michela Gallagher[4] & Karen H. Ashe[1,2,3,5]

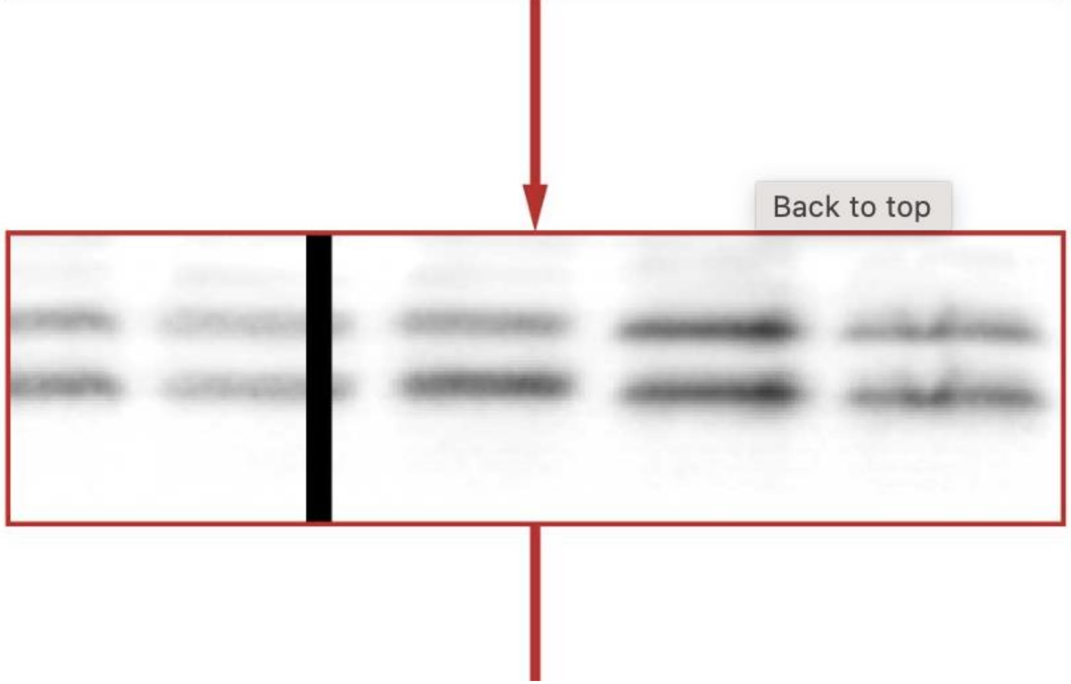
Image in question

Ashe uploaded this Western blot to PubPeer after Schrag said the version published in *Nature* showed cut marks suggesting improper tampering with bands portraying Aβ*56 and other proteins (black boxes added by Ashe). The figure shows levels of Aβ*56 (dashed red box) increasing in older mice as symptoms emerge. But Schrag's analysis suggests this version of the image contains improperly duplicated bands.



1 Spot the similarities

Some bands looked abnormally similar, an apparent manipulation that in some cases (not shown) could have made Aβ*56 appear more abundant than it was. One striking example (red box) ostensibly shows proteins said to emerge later in the life span than Aβ*56.



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> The immediate, obvious damage is wasted NIH funding and wasted thinking in the field because people are using these results as a starting point for their own experiments.

THOMAS SÜDHOF | STANFORD UNIVERSITY

Retraction Record Rocks Community

Anesthesiology tries to move on after fraud investigations

Breathe deeply: anaesthetists are well placed to conduct clinical trials with relatively little oversight.

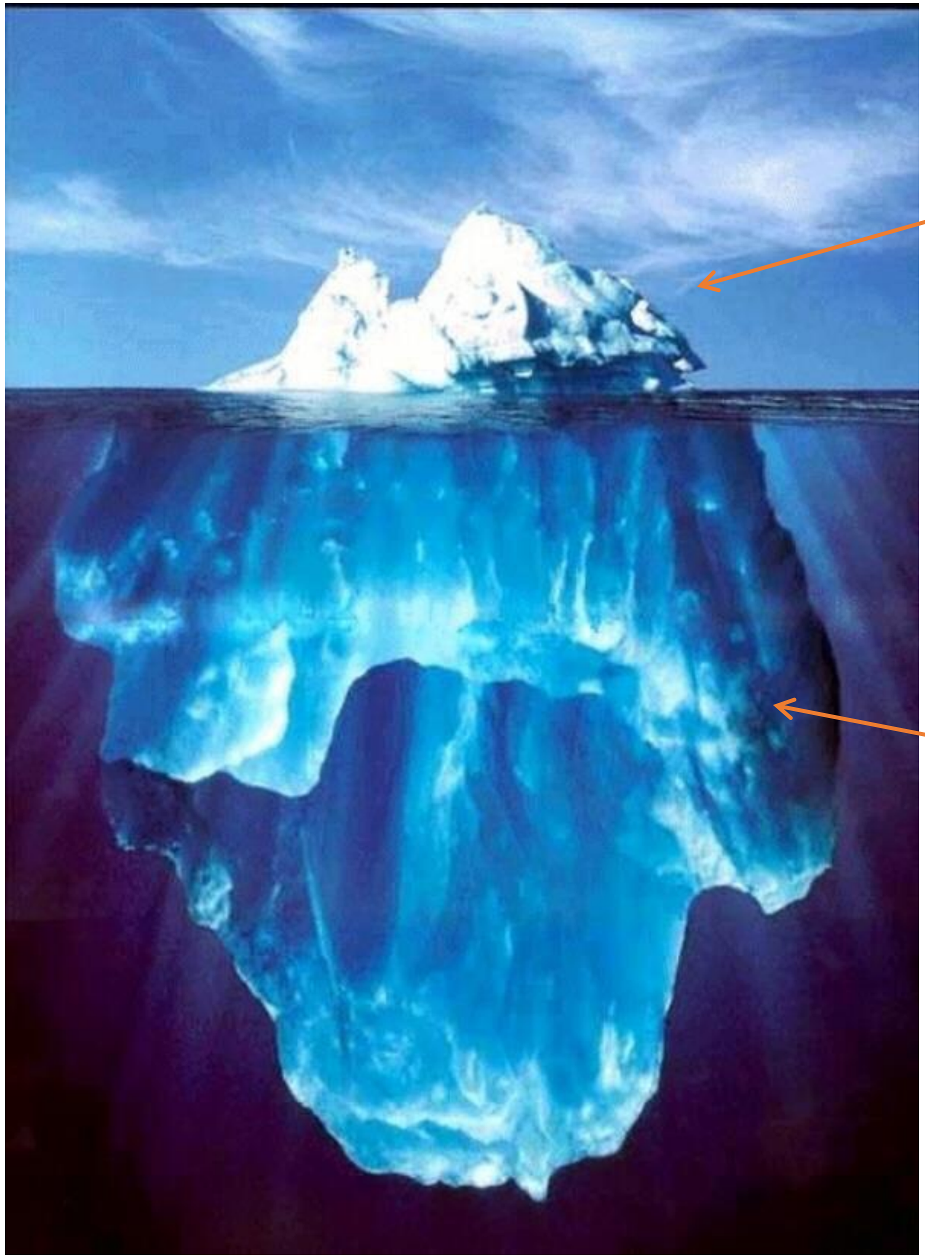
Nature (2012)

Fraud in Anesthesiology

- Scott Reuben, MD (1996 to 2010)
  - 21 fraudulent scientific articles retracted
  - Admits that he never conducted any of the trials he reported in the literature
  - Sentenced to prison (2010)

- Joachim Boldt, MD (1999 to 2012)
  - False data in 88 (of 102) scientific articles retracted

- Yoshitaka Fujii, MD (1993 to 2012)
  - 172 (of 212) fraudulent scientific articles, with 126 articles entirely fabricated
  - No ethical oversight, authorship inclusions


Data Fabrication/Falsification
Poor research conduct
(Conscious or Unconscious)

The Scandal of Poor Quality of Medical Research

- J Cohen (1962): Statistical power for medium effects in social science literature = 48%

- D Altman (2001):
  - Failure to specify eligibility criteria in 25% of surgery Journals
  - Failure to disclose the nature of blinding in 33%- 51% of RA journals, cystic fibrosis journals, and dermatology
  - Inadequate information on harmful consequences of interventions in 61% of Journals

Why Most Published Research Findings Are False

-Ioannidis (2005)

http://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.0020124

Exhibit A: Replication Crisis

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Replication crisis

From Wikipedia, the free encyclopedia

The **replication crisis** (or **replicability crisis**) refers to a methodological crisis in science in which scientists have found that the results of many scientific studies are difficult or impossible to replicate on subsequent investigation, either by independent researchers or by the original researchers themselves.[1] While the crisis has long-standing roots, the phrase was coined in the early 2010s as part of a growing awareness of the problem.

Since the reproducibility of experiments is an essential part of the scientific method, the inability to replicate the studies of others has potentially grave consequences for many fields of science in which significant theories are grounded on unreproduceable experimental work.

The replication crisis has been particularly widely discussed in the field of psychology (and in particular, social psychology) and in medicine, where a number of efforts have been made to re-investigate classic results, and to attempt to determine both the reliability of the results, and, if found to be unreliable, the reasons for the failure of replication.[2][3]

Fail to Replicate:	Others Work	Their Own Work
Chemistry	90%	60%
Biology	80%	60%
Physics	70%	50%
Medicine	70%	60%
Earth Science	60%	40%

Baker (2016). 1,500 scientists lift the lid on reproducibility

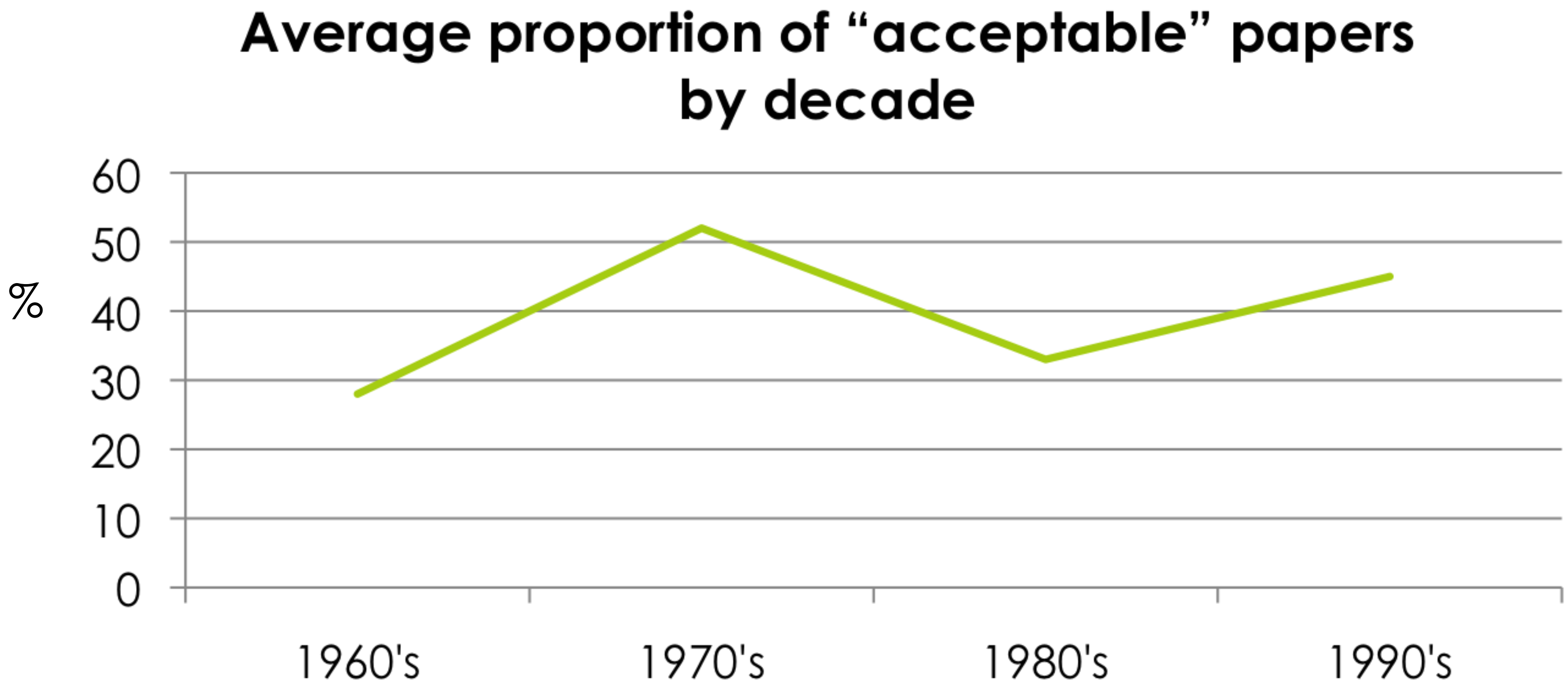
Causes of the Replication Crisis

- **Multifaceted**
  - Unprecedented rate of publication
  - Pressures to publish
  - Worship of novelty
  - Fraud
  - P-hacking
  - Researcher degrees of freedom
  - Selective Publication

The Problem with Statistical Reporting in Journals

- Incorrect choice of statistical techniques
- Flawed application of statistical methods
- Omission of crucial information to replicate study
  - Methods
  - Statistics
- Erroneous or biased conclusions
- Choice of statistical methods susceptible to bias or exaggerated claims that will fail to be replicated

How big of a problem is poor statistical analyses in medical journals?



Adapted from: Altman D. Statistical reviewing for medical journals. Statistics in Medicine, 1998; 17:2661 - 2674

What Journal (Statistical) Editors **Are Looking For**:

Methods, analyses, and results are reported with sufficient detail to allow replication

- Strictly adhere to a reporting guideline (e.g., CONSORT 2010)
  - 212 unique pieces of information in CONSORT
- Consider publishing/posting protocol at *start* of trial
- Develop and submit a statistical analysis plan (SAP) as part of the submission
  - JAMA template
  - Gamble et al. (2017). Guidelines for the content of statistical analysis plans in clinical trials
- Almost certainly requires supplemental content for completeness

Equator-network.org

Report Appropriate Bias Reduction Strategies

- Recruitment and sampling
- Thoughtful approach to randomization and allocation
  - Block sizes for multicenter studies
  - Stratification
  - Ratio may induce expectation effects (1:5 placebo to active drug)
- Blinding Methods
  - Who, how, where?
  - Threats to unblinding
- Intention to Treat Philosophy
  - Consideration of intercurrent events
  - Imputation or sensitivity analyses required?

Plan of Analysis is Summarized for Primary and Secondary Outcomes

- A priori power calculation with support for key assumptions
  - Poor assumptions make interpretation difficult
- Predefined versus post hoc analytical decisions clearly defined
- Exact plan of analysis (e.g., testing procedure) detailed for primary and secondary outcomes
- Type-I error strategy cleared defined and imposed on inferences

Ful SAP should be available as supplement or published/posted document

Provide Evidence of Expertise and Trustworthiness

- Ensure that all numerical findings are thoughtfully reported without error
  - Appropriate descriptive statistics
  - Contrasts and 95%CI (e.g., OR 2.2 [95%CI: 1.6 to 2.8], $p = 0.01$)
  - Triple check values in tables

- Avoid misuse of terms or methods
  - Blinding vs allocation concealment
  - Superiority vs equivalence
  - Change vs difference

Avoid Common Statistical Errors

- Misunderstanding the p-value
- Mistaking p-values for effect size
- Comparing p-values in lieu of interactions
- Overfitting models
- Post hoc subgroup analyses
- Artificial dichotomization of continuous variables
- Failing to consider assumptions
- Ignore missing data
- Unreliable measurements

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PLoS MEDICINE

Essay

Most Published Research Findings Are False—But a Little Replication Goes a Long Way

Ramal Moonesinghe*, Muin J. Khoury, A. Cecile J. W. Janssens

We know there is a lot of lack of replication in research findings, most notably in the field of genetic associations [1–3]. For example, a survey of 600 positive associations between gene variants and common diseases showed that out of 166 reported associations studied three or more times, only six were replicated consistently [4]. Lack of replication results from a number of factors such as publication bias, selection bias, Type I errors, population stratification (the mixture of individuals from heterogeneous genetic backgrounds), and lack of statistical power [5].

In a recent article in *PLoS Medicine*, John Ioannidis quantified the theoretical basis for lack of replication by deriving the positive predictive value (PPV) of the truth of a research finding on the basis of a combination